BRAC UNIVERSITY

Principles of Physics-II (PHY-112)

Department of Mathematics and Natural Sciences

Quiz: 01, **Section**: 02

Date: March 6, 2025

Duration: 30 Minutes | **Spring 2025 (10F-29C)** | Marks: 15

Use SI Units only. Read the questions properly before You start.

1. A positron and an electron enter an $\vec{E}$ field perpendicularly, and they both experience an acceleration that is proportional to their— **(1)**

   ◯ charge only ◯ charge-to-mass ratio and magnitude of $\vec{E}$ ✓ ◯ magnitude of $\vec{E}$ only ◯ mass only

   (Alternative) A proton and an electron enter an $\vec{E}$ field perpendicularly, and they both experience an acceleration that is proportional to their—

   ◯ charge only ◯ charge-to-mass ratio and magnitude of $\vec{E}$ ✓ ◯ magnitude of $\vec{E}$ only ◯ mass only

2. "Electric charges always come in fractions." This statement violates— **(1)**

   ◯ the conservation of charge principle ◯ the quantization of charge principle ✓ ◯ the additivity property of charges ◯ none of the above

   (Alternative) "Electric charges can never come in fractions." This statement describes—

   ◯ the conservation of charge principle ◯ the quantization of charge principle ✓ ◯ the additivity property of charges ◯ none of the above

3. A dipole, when immersed in a uniform electric field, starts to rotate in a direction decided by— **(1)**

   ◯ the potential energy it has stored in it ◯ the net torque it experiences ✓ ◯ its dipole moment vector only ◯ the net force it experiences

4. Two point charges placed r distance apart feels a mutual electrostatic force F_E. If both charges and the distance between them is doubled/tripled (Alternative), the new force is— **(1)**

   ◯ tripled ◯ cubed ◯ same ✓ ◯ one-third

5. What orientation (the angle between $\vec{p}$ and $\vec{E}$) of a dipole will give maximum torque and minimum electric potential energy for the system? **(1)**

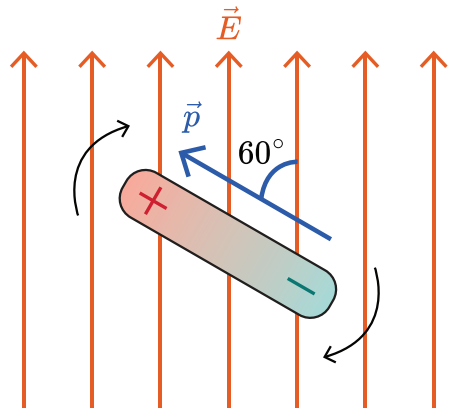
   ◯ 0° ◯ 90° ✓ ◯ 180° ◯ 270°

   (Alternative) What orientation (the angle between $\vec{p}$ and $\vec{E}$) of a dipole will give zero torque and maximum electric potential energy for the system? **(1)**

   ◯ 0° ◯ 90° ◯ 180° ✓ ◯ 270°

6. An electric dipole $\left(p = 5 \times 10^{-25}\,\mathrm{C\,m}\right)$ is placed from rest in a uniform electric field of magnitude $500\,\mathrm{N\,C^{-1}}$, directed upward. When placed, the dipole's $\vec{p}$ makes 60° with the field. Calculate the initial torque it experiences. In which direction? In or out of the page? [**Hint**: Drawing the diagram will help you find the torque's direction.] **(3)**

$\vec{\tau}$; into the page



The initial torque can be found using:

$$\begin{aligned}\tau &= pE\sin\theta \\ &= \left(5\times10^{-25}\,\mathrm{C\,m}\right)\cdot\left(500\,\mathrm{N\,C^{-1}}\right)\cdot\sin\left(60^\circ\right) \\ &= 2.165\times10^{-22}\,\mathrm{N\,m}.\end{aligned}$$

The rotation is clockwise since the torque decided by $\vec{p}\times\vec{E}$ is pointing into the page.

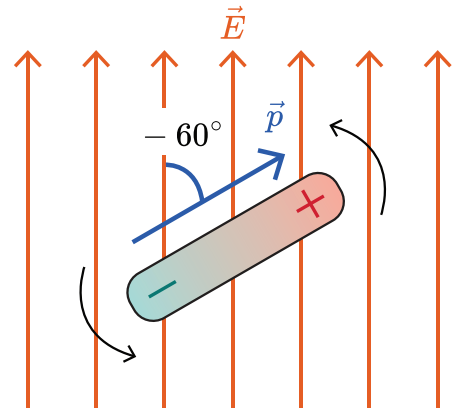
(Alternative) An electric dipole $\left(p = 5\times10^{-25}\,\mathrm{C\,m}\right)$ is placed from rest in a uniform electric field of magnitude $500\,\mathrm{N\,C^{-1}}$, directed upward. When placed, the dipole's $\vec{p}$ makes -60° with the field. Calculate the initial potential energy it gets stored in it. Which direction does the dipole rotate? Clockwise or anticlockwise? [**Hint**: Drawing the diagram will help you find the rotation direction.]

The initial potential energy stored can be found using:

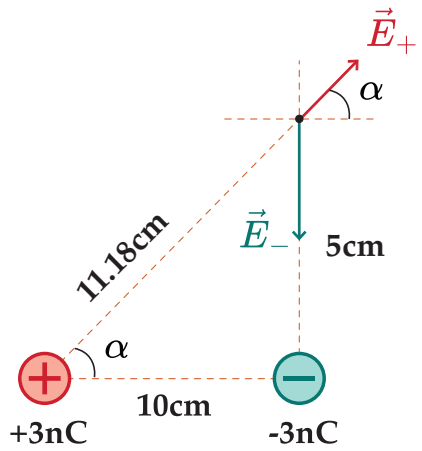
$$\begin{aligned}U_{\text{elec}} &= pE\cos\theta \\ &= \left(5\times10^{-25}\,\mathrm{C\,m}\right)\cdot\left(500\,\mathrm{N\,C^{-1}}\right)\cdot\cos\left(-60^\circ\right) \\ &= 1.25\times10^{-22}\,\mathrm{J}.\end{aligned}$$

The rotation is anticlockwise since the torque decided by $\vec{p}\times\vec{E}$ is pointing out of the page.

$\vec{\tau}$; out of the page



7. Calculate the horizontal and vertical components of the electric field produced by the two charges at the position indicated by the dot. Completing the diagram with vectors is mandatory. **(3)**



$$\alpha = \tan^{-1}\left|\frac{5}{10}\right| = 26.565^\circ$$

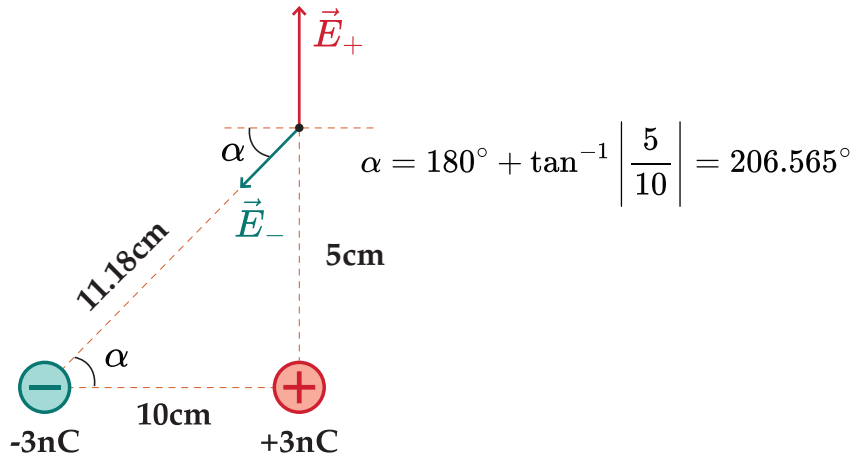
Calculate the fields produced by the two charges individually first:

$$\begin{aligned}\vec{E}_+ &= \frac{C|q_+|}{r_+^2}\left[\cos\{26.565^\circ\}\hat{i} + \sin\{26.565^\circ\}\hat{j}\right] \\ &= \frac{8.989\times10^9\,\mathrm{N\,m^2\,C^{-2}}\cdot 3\times10^{-9}\,\mathrm{C}}{\left(11.18\times10^{-2}\right)^2}\left[\cos\left(26.565^\circ\right)\hat{i} + \sin\left(26.565^\circ\right)\hat{j}\right]. \\ &= \left[1929.409\hat{i} + 964.702\hat{j}\right]\ \mathrm{N\,C^{-1}}.\end{aligned}$$

$$\vec{E}_- = \frac{C|q_-|}{r_-^2}\left(\cos 270^\circ\hat{i} + \sin 270^\circ\hat{j}\right)$$

$$= 0\hat{i} + \frac{8.989 \times 10^9\,\mathrm{N\,m^2\,C^{-2}} \cdot 3 \times 10^{-9}\,\mathrm{C}}{(5 \times 10^{-2})^2}\hat{j}.$$
$$= \left[-10.785 \times 10^3 \hat{j}\right]\ \mathrm{N\,C^{-1}}.$$

Horizontal component of the net field: $(1929.409 + 0)\ \mathrm{N\,C^{-1}} = 1929.409\,\mathrm{N\,C^{-1}}$.
Vertical component of the net field: $\left(964.702 - 10.785 \times 10^3\right)\ \mathrm{N\,C^{-1}} = -9820.298\,\mathrm{N\,C^{-1}}$.
(Alternative)



Calculate the fields produced by the two charges individually first:

$$\vec{E}_+ = \frac{C|q_+|}{r_+^2}\left[\cos\{90^\circ\}\hat{i} + \sin\{90^\circ\}\hat{j}\right]$$
$$= \frac{8.989 \times 10^9\,\mathrm{N\,m^2\,C^{-2}} \cdot 3 \times 10^{-9}\,\mathrm{C}}{(5 \times 10^{-2})^2}\left[\cos(90^\circ)\hat{i} + \sin(90^\circ)\hat{j}\right].$$
$$= \left[0\hat{i} + 10.785 \times 10^3 \hat{j}\right]\ \mathrm{N\,C^{-1}}.$$

$$\vec{E}_- = \frac{C|q_-|}{r_-^2}\left[\cos(180^\circ + 26.565^\circ)\,\hat{i} + \sin(180^\circ + 26.565^\circ)\,\hat{j}\right]$$
$$= \frac{8.989 \times 10^9\,\mathrm{N\,m^2\,C^{-2}} \cdot 3 \times 10^{-9}\,\mathrm{C}}{(5 \times 10^{-2})^2}\left[\cos(206.565^\circ)\hat{i} + \sin(206.565^\circ)\hat{j}\right].$$
$$= \left[-1929.7196\hat{i} - 964.8577\hat{j}\right]\ \mathrm{N\,C^{-1}}.$$

Horizontal component of the net field: $(0 - 1929.7196)\ \mathrm{N\,C^{-1}} = -1929.7196\,\mathrm{N\,C^{-1}}$.
Vertical component of the net field: $\left(10.785 \times 10^3 - 964.8577\right)\ \mathrm{N\,C^{-1}} = 9820.1423\,\mathrm{N\,C^{-1}}$.

8. Find the magnitude and direction of the dipole (formed by the two charges above figure)'s dipole moment vector in Question-7. **(2)**

The dipole moment vector $\vec{p}$ of the dipole in Question-7 can be found using:

$$\vec{p} = |q|\vec{d}$$
$$= \left(3 \times 10^{-9}\,\mathrm{C} \cdot 10 \times 10^{-2}\,\mathrm{m}\right)\left(-\hat{i}\right)$$
$$= \left(-3 \times 10^{-10}\hat{i}\right)\mathrm{C\,m}.$$

(Alternative)

$$\vec{p} = |q|\vec{d}$$
$$= \left(3 \times 10^{-9}\,\mathrm{C} \cdot 10 \times 10^{-2}\,\mathrm{m}\right)\hat{i}$$
$$= \left(3 \times 10^{-10}\hat{i}\right)\mathrm{C\,m}.$$

9. What equal amount of charge must be placed on both Mars and its Moon, Deimos, to neutralize their gravitational

attraction? [**Note**: $M_{\text{Mars}} = 6.39 \times 10^{23}\,\text{kg}$ and $M_{\text{Deimos}} = 1.51 \times 10^{15}\,\text{kg}$] **(2)**

$$
\begin{aligned}
F_G = F_E \qquad \Rightarrow G\frac{M_E M_M}{r^2} &= C\frac{Q^2}{r^2} \\
Q &= \sqrt{\frac{GM_{\text{Mars}}M_{\text{Deimos}}}{C}} \\
&= \sqrt{\frac{6.674 \times 10^{-11}\,\text{Nm}^2\text{kg}^{-2} \times 6.39 \times 10^{23}\,\text{kg} \times 1.51 \times 10^{15}\,\text{kg}}{8.99 \times 10^{9}\,\text{Nm}^2\text{C}^{-2}}} \\
&\approx 2.676 \times 10^{9}\,\text{C}.
\end{aligned}
$$

(Alternative) What equal amount of charge must be placed on both Saturn and its Moon, Titan, to neutralize their gravitational attraction? [**Note**: $M_{\text{Saturn}} = 5.683 \times 10^{26}\,\text{kg}$ and $M_{\text{Titan}} = 1.345 \times 10^{23}\,\text{kg}$]

$$
\begin{aligned}
Q = \sqrt{\frac{GM_{\text{Saturn}}M_{\text{Titan}}}{C}} &= \sqrt{\frac{6.674 \times 10^{-11}\,\text{Nm}^2\text{kg}^{-2} \times 5.683 \times 10^{26}\,\text{kg} \times 1.345 \times 10^{23}\,\text{kg}}{8.99 \times 10^{9}\,\text{Nm}^2\text{C}^{-2}}} \\
&\approx 7.5329 \times 10^{14}\,\text{C}.
\end{aligned}
$$